

Module 12: Installation, Storage, and Compute with Windows Server

Section 1: Installation & Configuration

1. What two options are provided in the type of installation window during Windows Server 2016 installation?

- **Upgrade:** Installs Windows and keeps files, settings, and applications.
- **Custom:** Installs Windows only (Clean Installation). This is used for new hard drives or starting from scratch.

2. Write the step: How to configure a server step-by-step?

1. Assign a **Static IP address** to the network interface.
2. Change the **Computer Name** to a standard naming convention.
3. Join the server to a **Domain** (if not becoming a DC).
4. Configure **Windows Updates** to ensure the latest security patches.
5. Enable **Remote Desktop** for remote management.
6. Install necessary **Roles and Features** via Server Manager.

3. What are the Pre-installation tasks?

- Check hardware compatibility (HCL).
- Disconnect UPS devices.
- Back up existing data if performing an upgrade.
- Ensure you have the correct Product Key and Installation Media.
- Disable Antivirus software (if upgrading).

4. What are the Post-installation tasks?

- Activate Windows Server.
- Install the latest hardware drivers.
- Configure Windows Firewall.
- Set the correct Date, Time, and Time Zone.
- Install Management Tools (RSAT).

5. What is the standard upgrade path for Windows Server?

Microsoft generally supports an "N-2" upgrade path. For example, you can upgrade directly from Windows Server 2012 R2 to 2016, or 2016 to 2019/2022. You cannot skip more than two versions without a clean install.

Section 2: Active Directory Fundamentals

6. What is the Physical structure of AD?

The physical structure consists of **Domain Controllers** (servers that hold the database) and **Sites** (which represent the physical network topology/locations and manage replication traffic).

7. What are the Logical components of Active Directory?

- **Partitions**
- **Schemas**
- **Domains**
- **Trees**
- **Forests**
- **Organizational Units (OUs)**

8. What is the Full form of LDAP?

Lightweight Directory Access Protocol.

9. What is the location of the AD database?

The database file (**ntds.dit**) is located in %SystemRoot%\NTDS (usually C:\Windows\NTDS).

10. What is a Child DC?

A Child Domain Controller is a domain that shares a contiguous namespace with its parent domain but maintains its own separate database and administrative boundaries within the same Forest.

11. Explain the term "Forest" in AD.

A Forest is the highest level of the logical container. It is a collection of one or more domain trees that share a common Schema, Configuration, and Global Catalog.

Section 3: Assignment MCQ & Short Answer

12. What is Active Directory? (Check all that apply)

- ☒ Microsoft's implementation of a directory server
- ☒ An LDAP-compatible directory server

13. When you create an Active Directory domain, what's the name of the default user account?

- **Administrator**

14. AD domain provides which of the following advantages? (Check all that apply)

- [X] Centralized authentication
- [X] Centralized management with GPOs

15. What are the minimum hardware requirements for installing Windows Server 2016?

- **Processor:** 1.4 GHz 64-bit processor.
- **RAM:** 512 MB (2 GB for Server with Desktop Experience).
- **Disk Space:** 32 GB.

16. Explain the different editions of Windows Server 2016 and their features.

- **Datacenter:** For highly virtualized environments; includes Shielded VMs and Storage Spaces Direct.
- **Standard:** For physical or lightly virtualized environments (includes 2 Hyper-V containers).
- **Essentials:** For small businesses (up to 25 users/50 devices).

17. Walk through the steps of installing Windows Server 2016 using GUI mode.

1. Boot from installation media.
2. Select Language, Time, and Keyboard. Click **Install Now**.
3. Enter Product Key or select "I don't have a key."
4. Select **"Windows Server 2016 (Desktop Experience)"**.
5. Accept License terms.
6. Choose **Custom Installation**.
7. Select the partition and click **Next** to begin installation.

18. Describe the steps for installing Windows Server 2016 in Server Core mode.

The steps are identical to the GUI installation, except in Step 4, you select **"Windows Server 2016"** (the option *without* Desktop Experience). Once installed, you manage it via Command Prompt or PowerShell.

19. How do you configure network settings during Windows Server 2016 installation?

After the first login, use **Sconfig** (for Server Core) or **Network and Sharing Center** (for GUI) to set a static IP, Subnet Mask, Default Gateway, and DNS Server.

20. Explain the process of promoting a Windows Server to a domain controller.

1. Open **Server Manager** -> Add Roles and Features.
2. Install **Active Directory Domain Services (AD DS)**.
3. Click the notification flag and select **"Promote this server to a domain controller"**.
4. Choose to "Add a new forest" and specify the Root Domain Name.

5. Set a DSRM password and complete the wizard.

21. Discuss the steps involved in upgrading from a previous version to Windows Server 2016.

1. Verify the path (e.g., 2012 R2 -> 2016).
2. Run Setup from within the old OS.
3. Select **"Upgrade: Install Windows and keep files, settings, and apps"**.
4. Wait for the compatibility check and proceed with the automated installation.

22. What is Active Directory Domain Services (AD DS), and what are its key components?

AD DS is a server role that stores information about network objects and makes this information easy for administrators and users to find and use. Key components include the **Data Store (ntds.dit)**, **Domain Controllers**, and the **Global Catalog**.

23. How do you create a new Active Directory user account in Windows Server?

Open **Active Directory Users and Computers (ADUC)**, right-click an OU or the Users container, select **New -> User**, and fill in the name and login credentials.

24. Explain the process of creating and managing Group Policy Objects (GPOs).

Open **Group Policy Management Console (GPMC)**, right-click "Group Policy Objects," select **New**, and give it a name. Right-click the new GPO and select **Edit** to change settings. Finally, link the GPO to a specific Domain or OU.

25. What are Organizational Units (OUs) in Active Directory, and how do you use them?

OUs are containers within a domain used to organize objects (users, computers). They are primarily used to simplify administration and to **link Group Policies** to specific groups of users or machines.

26. Describe the process of delegating administrative privileges in Active Directory.

Right-click an OU in ADUC and select **Delegate Control**. Follow the wizard to choose a user/group and assign specific permissions (like "Reset User Passwords") without giving them full domain admin rights.